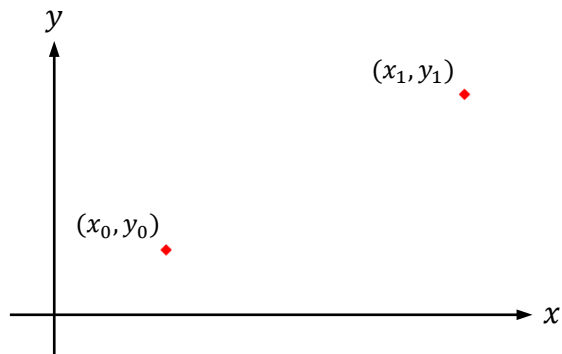


Newton Interpolating Polynomials



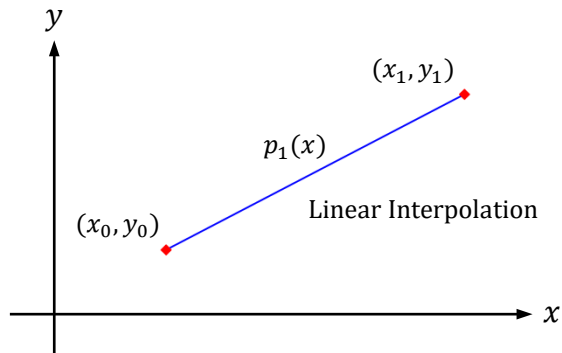
Linear Interpolation

$$p_1(x) = C_0 + C_1 \cdot (x - x_0); \quad x_0 \leq x \leq x_1$$

$$p_1(x_0) = f(x_0): \quad f(x_0) = C_0 + C_1 \cdot (x_0 - x_0) \quad \rightarrow \quad C_0 = f(x_0)$$

$$p_1(x_1) = f(x_1): \quad f(x_1) = C_0 + C_1 \cdot (x_1 - x_0)$$

$$f(x_1) = f(x_0) + C_1 \cdot (x_1 - x_0) \quad \rightarrow \quad C_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$



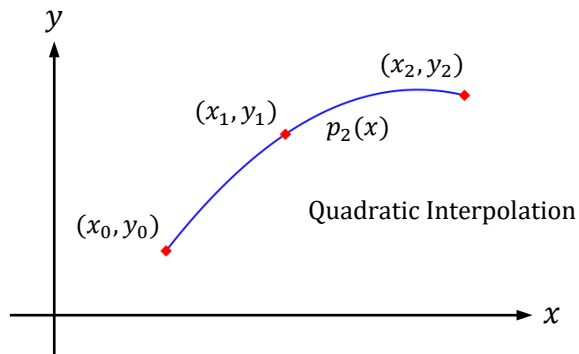
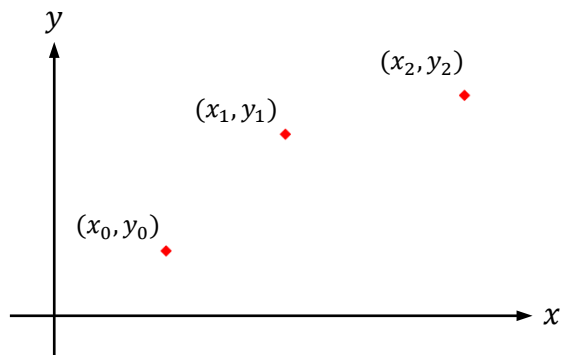
For $x_0 \leq x \leq x_1$,

$$p_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} \cdot (x - x_0)$$

$$p_1(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0)$$

$$f[x_0] = f(x_0), f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Newton Interpolating Polynomials



Quadratic Interpolation

$$p_2(x) = C_0 + C_1 \cdot (x - x_0) + C_2 \cdot (x - x_0) \cdot (x - x_1); \quad x_0 \leq x \leq x_2$$

$$p_2(x_0) = f(x_0): \quad f(x_0) = C_0$$

$$p_2(x_1) = f(x_1): \quad f(x_1) = C_0 + C_1 \cdot (x_1 - x_0)$$

$$p_2(x_2) = f(x_2): \quad f(x_2) = C_0 + C_1 \cdot (x_2 - x_0) + C_2 \cdot (x_2 - x_0) \cdot (x_2 - x_1)$$

$$f(x_2) = C_0 + C_1 \cdot (x_2 - x_1 + x_1 - x_0) + C_2 \cdot (x_2 - x_0) \cdot (x_2 - x_1)$$

$$f(x_2) = f(x_1) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} \cdot (x_2 - x_1) + C_2 \cdot (x_2 - x_0) \cdot (x_2 - x_1)$$

For $x_0 \leq x \leq x_2$,

$$p_2(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} \cdot (x - x_0) + \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0} \cdot (x_2 - x_0) \cdot (x_2 - x_1)$$

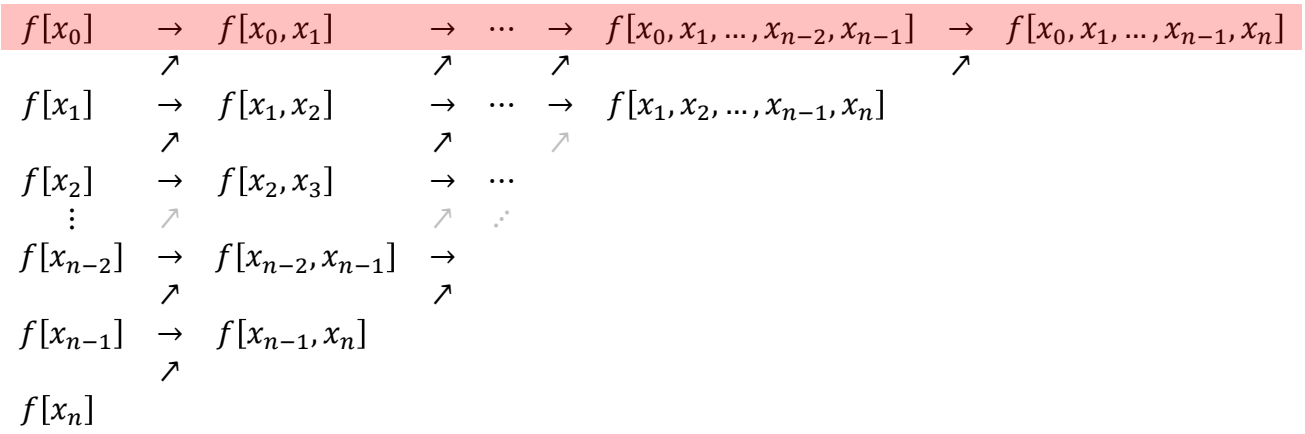
$$p_2(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1)$$

Newton Interpolating Polynomials

General Form

$$p_n(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1) + \dots + f[x_0, x_1, \dots, x_n] \cdot (x - x_0) \cdot (x - x_1) \cdots (x - x_{n-1}); \quad x_0 \leq x \leq x_n$$

$$f[x_0] = f(x_0), f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}, f[x_0, x_1, x_2] = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0} = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}, \dots$$



Newton Interpolating Polynomials

Example 04-01a: Approximate $f(0.45)$ from the following data using Newton polynomials.

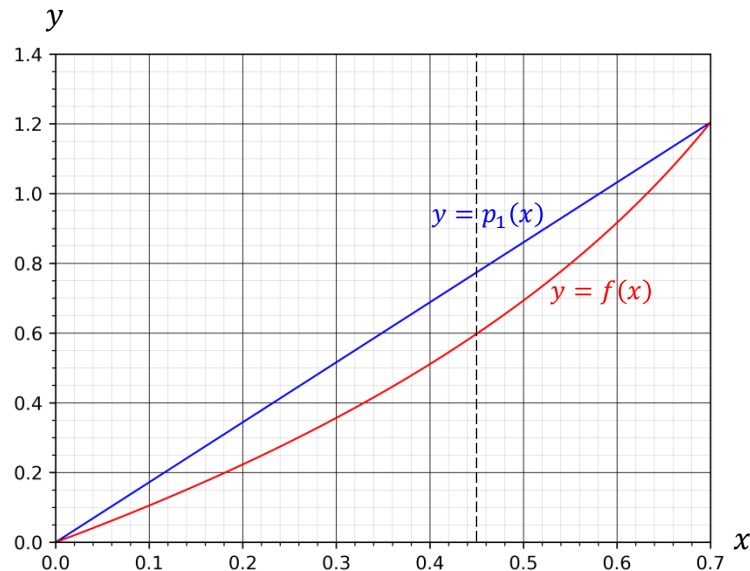
i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728

$$\begin{aligned}p_1(x) &= f[x_0] + f[x_0, x_1] \cdot (x - x_0) \\&= 0.0000000 + 1.7199611 \cdot (x - 0.0)\end{aligned}$$

$$\begin{aligned}f(0.45) \approx p_1(0.45) &= 0.0000000 + 1.7199611 \cdot (0.45 - 0.0) \\&= 0.7739825\end{aligned}$$

$$f(x) = -\ln(1 - x)$$

$$f(0.45) = -\ln(1 - 0.45) = 0.5978370$$



Newton Interpolating Polynomials

Example 04-01b: Approximate $f(0.45)$ from the following data using Newton polynomials.

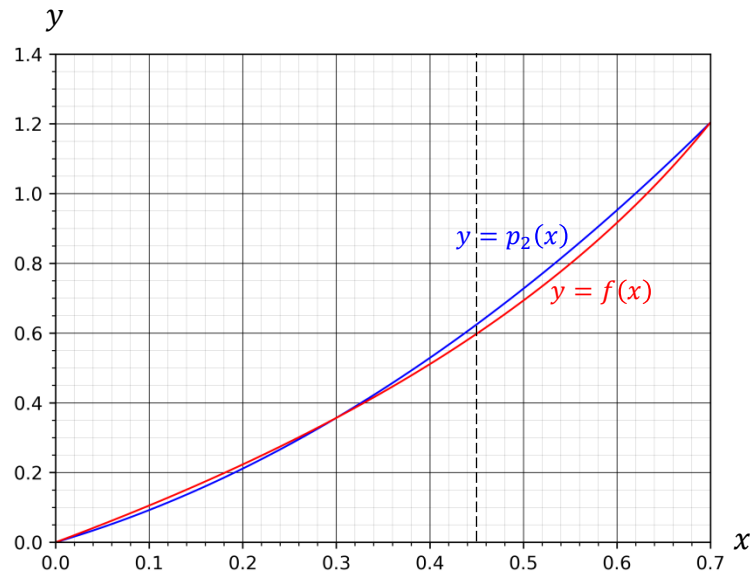
i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749

$$\begin{aligned} p_2(x) &= f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1) \\ &= 0.0000000 + 1.7199611 \cdot (x - 0.0) + 1.3276120 \cdot (x - x_0) \cdot (x - x_1) \end{aligned}$$

$$\begin{aligned} f(0.45) \approx p_2(0.45) &= 0.0000000 + 1.7199611 \cdot (0.45 - 0.0) \\ &\quad + 1.3276120 \cdot (0.45 - 0.0) \cdot (0.45 - 0.7) \\ &= 0.6246262 \end{aligned}$$

$$f(x) = -\ln(1 - x)$$

$$f(0.45) = -\ln(1 - 0.45) = 0.5978370$$



Newton Interpolating Polynomials

Example 04-01c: Approximate $f(0.45)$ from the following data using Newton polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

$$p_3(x) = f[x_0] + f[x_0, x_1] \cdot (x - x_0) + f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1) \\ + f[x_0, x_1, x_2, x_3] \cdot (x - x_0) \cdot (x - x_1) \cdot (x - x_2)$$

$$f(0.45) \approx p_3(0.45) = 0.6045237$$

$$f(x) = -\ln(1 - x)$$

$$f(0.45) = -\ln(1 - 0.45) = 0.5978370$$

